Practical 8

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3)

Code:

```
  GNU nano 8.7
#include<stdio.h>

int main()
{
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3];
    int i,j,k,found;
    int hit=0,fault=0;

    for(i=0;i<3;i++)
        frames[i] = -1;

    for(i=0;i<12;i++)
    {
        found = 0;

        for(j=0;j<3;j++)
        {
            if(frames[j] == pages[i])
            {
                hit++;
                found = 1;
                break;
            }
        }

        if(found == 0)
        {
            frames[i%3] = pages[i];
            fault++;
        }
    }

    printf("Page Hits = %d\n", hit);
    printf("Page Faults = %d\n", fault);

    printf("Hit Percentage = %.2f\n",(hit/12.0)*100);
    printf("Fault Percentage = %.2f\n",(fault/12.0)*100);

    return 0;
}
```

Output:

```
HP@LAPTOP-9M3IBDT5 MSYS ~
$ cd

HP@LAPTOP-9M3IBDT5 MSYS ~
$ nano list.c

HP@LAPTOP-9M3IBDT5 MSYS ~
$ gcc list.c

HP@LAPTOP-9M3IBDT5 MSYS ~
$ ./list
-bash: ./list: No such file or directory

HP@LAPTOP-9M3IBDT5 MSYS ~
$ ./a.exe
Page Hits = 2
Page Faults = 10
Hit Percentage = 16.67
Fault Percentage = 83.33

HP@LAPTOP-9M3IBDT5 MSYS ~
$ |
```